Every Door a Verdict: the Closure Record of the a_0 Galaxy Law and its Screened Khronon Action

Working note (PAPER69), version 1.1, 2026-11-06 (v1.0: DOI 10.5281/zenodo.24025170). AI-assisted research programme; not peer reviewed. Companions: PAPER67 (DOI 10.5281/zenodo.23968171, the law and its failure record) and PAPER68 (DOI 10.5281/zenodo.23990944, the action). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

"Closure" here means that every gate the programme set itself has a frozen verdict, not that every gate passes. On 2026-11-06 the programme ran each remaining gate for the galaxy law $g_N = g^2/(g + a_0)$, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, and for its adopted, scoped action (v3s; PAPER68) and candidate successor (AC2q). Each lane froze its criteria before any number was computed and ran a mutation twin that must fail. The full covariant field equations of both actions are now derived and checked against direct variation. Every step from the action to the observables is tagged as Lean-proved, symbolic, numerical or postulated. The law passes SPARC on the action's exact field on both footings, along with the Solar System, PPN, the tensor speed and galaxy lensing. It fails clusters, CMB lensing and linear growth. These are no-go results over every term the action has, not missed fits. Among galaxies it fails the X-ray ellipticals, two ultra-diffuse galaxies (Malin 1, AGC 114905) and, on the canonical footing only, the Υ-tilt row. The failures form a single step: hot, pressure-supported systems sit 0.32–0.37 dex above cold discs at matched baryonic mass, acceleration, depth, radius and gas fraction. Five candidate keys for that step were each frozen and graded: environment, cold gas, kinematic support, morphology and IMF. None explains it; IMF variation removes about half of it in ellipticals only. Version 1.1 adds three further mechanisms built to pass every gate (a power-law-tail sector, a two-phase superfluid condensate, a dipolar medium); all fail. It also adds two audits of the ΛCDM assumptions inside the cluster and growth data; both find the failures robust. We list the gates that wait on data (Gaia DR4, a proposal-class ALMA sample, the unscored ALMA release rows) and the decisions that wait on the owner.

1 What "closure" means here

A gate is closed when it carries one of four labels, each fixed by criteria committed before the first graded number: PASS, FAIL, a proven no-go (FAIL for a whole class), or CONDITIONAL with the condition named. NO VERDICT closes a gate only when the lane shows the test has no power, and says what data would give it power. A gate that is RUNNING is open. Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are carried throughout and never pooled. κ is fitted; a Lean proof shows that it is not derivable from the programme's premises.

2 The actions on the table

v3s, adopted and scoped (PAPER68). This is a Blanchet–Skordis Khronon theory with a postulated acceleration function and a switched stiffness $-(\gamma_L/2)[\max(D\cdot P, 0)]^2$. QX17 finds a misprint in the action as printed in PAPER68: read literally, it decouples P. The form every lane actually solved puts $\mathcal{J}$ in Legendre form, $P\cdot A - W(K|P|)$. QX17 derives the complete covariant Euler–Lagrange set from that operative form and checks it against direct sympy variation. That is the erratum PAPER68 needs.

AC2q, candidate, not adopted. The dark-energy scalar is the clock. The switch acts on the khronon charge $\Sigma = D{\cdot}P + 2X$, and there is no dust term (QX17A, field equations CERTIFIED). Its FRW lapse factor is $(1-\gamma\dot\phi^2)^2$, not $(1-2\gamma\dot\phi^2)^2$ (FRW reconciliation chain; errata 0cbca2a4f2). For any $f \geq 0$ the signs $0 < c_g \leq c_{\rm ph} \leq 1$ hold (Lean).

Repair, not adopted. Replace the kink $\max(\cdot, 0)^2$ with AC21's blend: a pure Π^6 near zero, C^4-blended to $(\gamma/2)\Pi^2$ above $b = \rho_{\rm DE}$ (v3s) or $b = X$ (AC2q). It adds no constant.

3 The equation chain

EQUATIONS_IN_ORDER.md (QX1) runs action → field equations → FRW → static law → PPN → c_T → lensing → Cassini/$L_\star$ → AC2q reductions. Each step is tagged LEAN, SYMPY, NUM or POSTULATED. QX1 carries 28 Lean theorems with 0 `sorry` and standard axioms only. The lane files add more, each with a MUTATE twin that fails on its false claims. The FRW reconciliation chain (closure_2026_11_06/FRW_RECONCILIATION_CHAIN/CHAIN.md, 8d075bf20f) re-derives the AC2q scalar FRW reduction independently by ADM. Lean `FRW_correction` has 11 theorems; its MUTATE twin fails 3 of 3. With the corrected coefficient, AC23 PASS reproduces (max R 2.700110 against 2.700138), and the AC13 stress rows at 0.81 give R 1.000 to 2.5257 against the stored 1.000 to 2.5253. These are post-hoc reruns, not frozen regrades. QX17B (c3cebe6cd6) is NOT CERTIFIED as a whole. Its Lean kernel certificate passes (12 theorems; the MUTATE twin fails M1–M4), the ADM form with c restored is written, and the galaxy law plus GR remainder pass on both footings. The spherical solver did not converge on the known case or on the full nonlinear Sun; that is a solver failure, not physics. The 3+1 jet variation was not run (3 h cap), and the numerical ADM check is being rerun after an output-path bug.

4 Verdicts

Gate	Verdict	Lane
Pass		
SPARC/RAR, exact AQUAL field	PASS both footings: rms 0.0560 at Υ 0.58 (canonical), 0.0558 at 0.52 (alt)	QX8
Solar System, PPN, c_T, galaxy lensing	PASS (v3s, carried; AC2q tensor and slip)	PAPER68, AC9
PN sign, all directions	PASS, scoped (UV principal symbol)	AC17
AC2q slow roots, DE-like backgrounds	PASS (harmless, scoped), $R \leq 2.70$	AC23 + FRW rerun
Field equations, v3s and AC2q	CERTIFIED vs direct variation	QX17, QX17A
One-action scorecard (AC2q + Π^6)	all ten carries CARRY; now CONDITIONAL (gradient catastrophe if a region sits in $\Sigma > X$)	QX19, errata
Well-posedness, Π^6 repair	3D condition proved; curved 2D C^1, C^2 PASS	AC21, AC22
Fail, no-go over the whole class		
Clusters, every static term	FAIL: no universal ν; X-COP plain law $+0.468/+0.435$	QX2
Clusters, keyed and derived terms	FAIL (pressure, khronon velocity, density, gas-g; depth key not derivable)	QX5, QX6
Hot sectors (shock-heated, isothermal)	FAIL for every parameter	QX12, QX13
CMB lensing and growth	FAIL, no-go (clock Jeans; $K(Q)$ short; barotropic class)	QX3
Switched cold-dark-energy window	KILLED (galaxies retain the sector)	QX7, QX10
Primary CMB (PR4 CamSpec + ACT DR6)	CONDITIONAL on a cold component: with it, PASS both footings (χ^2 3741.08/3741.15, $\Delta = -64.7$ against the profiled yardstick; h 0.674 fixed by a_0). Baryons only: FAIL by +172,848/+166,612. Not independent evidence (fixed-h background plus a cold component gives the standard spectrum), and the record kills every cold sector in galaxies (row above)	QX15
Well-posedness, raw switch	FAIL across the switch for generic data	AC16
Constraint causality (v3s, AC2q)	Light-cone causality FAIL: a conserved source changes the gauge-invariant $R_{\mu\nu}u^\mu u^\nu$ outside its light cone through the leaf-elliptic mode on a charged anisotropic background (all 36 graded observer rows nonzero, both footings; $\lvert\delta v\rvert$ from 3.6×10^{-67} (canonical) to 7.8×10^{-66} m/s (alt) for 1000 kg moved at 1 m/s, seen 1 s later at 10^9 m). Causal ordering along the foliation PASS, scoped. Isotropic and zero-charge controls are causal	AC18

Gate	Verdict	Lane
Fail, galaxies		
X-ray ellipticals, G7 line 0.15	FAIL: $D = +0.210/+0.189$	QX18
Υ-tilt row	FAIL canonical +0.170, PASS alt +0.090	QX11
Malin 1	FAIL, class no-go	QX4
AGC 114905	NOT RESCUED: no external-field host in the census	QX14
FN6 window source	FAIL for v3s/AC2q	AC19
Early black holes (A2744-QSO1 by z 7.05)	FAIL: in WG2's own law-on external field no baryon clump 10^6–$10^{13}\,M_\odot$ collapses before z 0.48 (canonical, weakest p05 field) or ever (median field); EB1's isolated-sphere EASY was an artefact. Growth no-go at seed scales	EB1, EB1b
The step and its candidate keys		
Failure ladder	NO SINGLE VARIABLE: hot systems +0.32–0.37 dex above SPARC at matched variable	QX20
Weak-lensing early/late split	NO VERDICT: $+0.158 \pm 0.064$ (2.46σ), same sign	QX21
Kinematic-support key	NO VERDICT (narrow), killed under all three mappings	QX22
Cold-gas key / baryon gap	NEITHER: both excluded	QX23
Environment key	KILLED as frozen (fragile on the max-cluster reading)	QX24
Morphology key	NO VERDICT inside SPARC ($+0.078 \pm 0.045$); colour and Sérsic split equally in KiDS	QX25
IMF variation (Posacki+2015 $\alpha(\sigma_e)$)	PARTLY for early types: X-ray ellipticals $+0.210 \rightarrow +0.093$ (canonical), inside the G7 line, with 56–61 % removed against 0.8 needed; KiDS early-type residual +0.080 to +0.122. NOT for X-COP (4 % removed), groups or dSphs. SPARC pass unchanged	QX26 8949f793af (criteria 44b5658218)
No verdict as frozen		
Nonlinear fold, full grid (v3s, AC2q)	NO VERDICT as frozen: the rule gives NO VERDICT if any physical row is UNDECIDED, and 296 of the 485 parts are UNDECIDED on solver stops before a graded event (every AC2q clock-only and phantom row, every v3s phantom row, 56 of 80 AC2q soft rows). Every v3s clock-only and soft row folds. The explicit integrator stops at the first edge fold, and one MUTATE control does not bite	AC14 bde9320c80 (criteria ad9d747016)
Moving Sun, powered control	NO VERDICT as frozen: power and order checks PASS, but medium→fine convergence FAILS (0.496/0.572 against 0.30), all from the q_B Bernoulli proxy, which falls under refinement. Worst over every medium bracket: Mars 0.20 of its line, Cassini 0.028. Post-data reading PASS on q	AC24
Wait on data		
$a_0(z)$	needs a proposal-class ALMA sample	AT2
Wide binaries	Gaia DR4 (pre-registration frozen)	DR4
ALMA 2026-11-06 release, rows B1–B4	UNSCORED: row-B fits collapse to $I_0 = 0$ on real data (AL1, AL2); addenda G3 fails and G4 is not validated on mocks (AL3, AL4). Row A upper limit as predicted	AL1–AL4

5 What the record says, plainly

1. The law, and the action in its scope, describe cold rotating discs (SPARC on the exact field, both footings) and the Solar System. The exceptions among discs and dwarfs are the Υ-tilt row on the canonical footing and two ultra-diffuse galaxies. The action's health failures (light-cone causality, the raw switch) are structural; the first is far below any measurement.
2. Outside discs the law misses by a *step*, not a trend. No measured present-day variable we tried keys it. Environment, cold gas, support and morphology are each killed or give no verdict. A heavier IMF in high-σ spheroids (QX26) removes about half of the step in ellipticals and almost none in groups, clusters or dSphs, so the step that remains is carried by the systems dominated by hot gas.
3. The action has no term that can supply cluster mass or CMB-lensing growth. Those are proven no-gos over the action's whole term class, so a fix needs new physics, not a refit.

4. An idea checked after EB1, that early-collapse black holes are the hot systems' missing mass, fails on numbers already on the record. The per-star yield would have to differ by at least 15× between ellipticals and clusters, and a gas-proportional amount fails the Bullet (closure_2026_11_06/IDEA_EARLY_BH_DARK_BARYONS_CHECK.md).

6 Version 1.1: every remaining door, and the data's own assumptions

After v1.0 the owner asked for a mechanism that passes every gate. Three classes were built as test lanes, each with its constants fixed from galaxies, KiDS and dSphs only (no cluster number), on both footings of this paper's a_0 (never the standard-MOND value). Two lanes then audited the ΛCDM- and simulation-dependent steps inside the data products the failures rest on.

- **Power-law-tail sector** (QX27 497936beb0): cold before $z = 10$, then heated into a v^{-n} tail so retention rises smoothly with escape speed. Its statics are the best no-input cluster result in the record (X-COP $-0.024/0.085$ canonical at $n = 4$), but it FAILS growth and the forest ($\Delta\chi^2$ 231–1059 against 4) and phase-space capture (clusters need 5.6×10^3–7.0×10^3 times the Liouville bound). A sector hot enough to stay out of galaxies neither clusters nor can be captured.
- **Two-phase superfluid condensate** (QX28 9d4b9dae31; Berezhiani–Khoury class, test lane only): galaxies pass, and KiDS is not over-predicted because the phonons are sourced by baryons alone. It FAILS clusters (normal phase X-COP +0.120/rms 0.139; fully condensed overshoots), ellipticals, groups, growth ($\Delta\chi^2$ 140/70 against 4) and Cassini.
- **Dipolar polarizable medium** (QX29 ff71610edf; test lane only): reproduces the law exactly in galaxies, and FAILS Cassini for every host rule (Q2 2.44–4.55× the Hees line); every placement of its monopole loses one corner of the triangle (galaxy-calibrated amplitude leaves X-COP at +0.389).
- **Cluster-data assumptions** (QX30 e9e9fc34b6): ROBUST. The one admissible framework-native correction, each paper's fiducial D_A replaced by the framework's own, moves X-COP by -0.0205 dex (to $+0.447/+0.414$). Non-thermal pressure and hydrostatic bias make the miss larger. Every favourable bound at once still leaves X-COP at $+0.329/+0.298$.
- **Growth and CMB-data assumptions** (QX31 636774e806): ROBUST. No verdict moves. The fiducial corrections make the law-alone CMB-lensing misses worse; even a non-admissible generous bound leaves the shape χ^2 at 370–399 against 20.09, and all CamSpec nuisance freedom is 0.4 % of the baryons-only primary-CMB gap.

The data were not hiding an answer. Every mechanism that keeps the law in galaxies either enters galaxy halos as lensing mass, cannot grow structure, or cannot be captured by clusters.

7 What waits on the owner

Adoption of the Π^6 stiffness repair; the PAPER68 erratum for the Legendre form (QX17); the KiDS catalogue at fixed stellar mass (QX25 NEEDS_OWNER).

Method

Every lane froze its criteria in a commit before any script ran. Every lane has a MUTATE twin that must fail, and every rerun was byte-identical where stated. Footings are reported apart. No ΛCDM inputs were used. Scripts and outputs are in the repository at the commits the lane READMEs name.